The Top 10 Common Methodological and Statistical Mistakes in Anesthesiology Research

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Ten topics that are commonly seen in the Anesthesiology literature

- Gathered from numerous articles, and my own experience in evaluating studies

- The topics are issues related to common (frequentist) methods; only brief mention of other methods

- The 10 topics will be presented in a rank order of importance
  - Added drama
  - Prevent boredom from killing you

10

Lack of detail about the analytical plan

- Provide enough detail to allow replication if allowed access to your data:
  - Statistical Software
  - Criterion for statistical significance ($p < 0.05$)
  - Hypothesis testing (two-tailed)
  - Assumptions underlying tests (normality, etc.)
  - Actual applied tests, not just a list
  - Reporting of data (mean, median, etc.)
  - Correction for multiplicity

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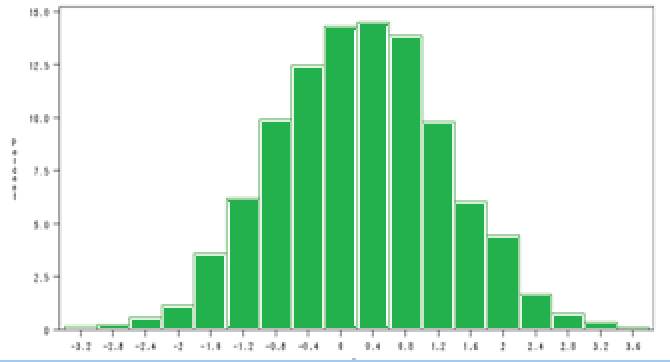
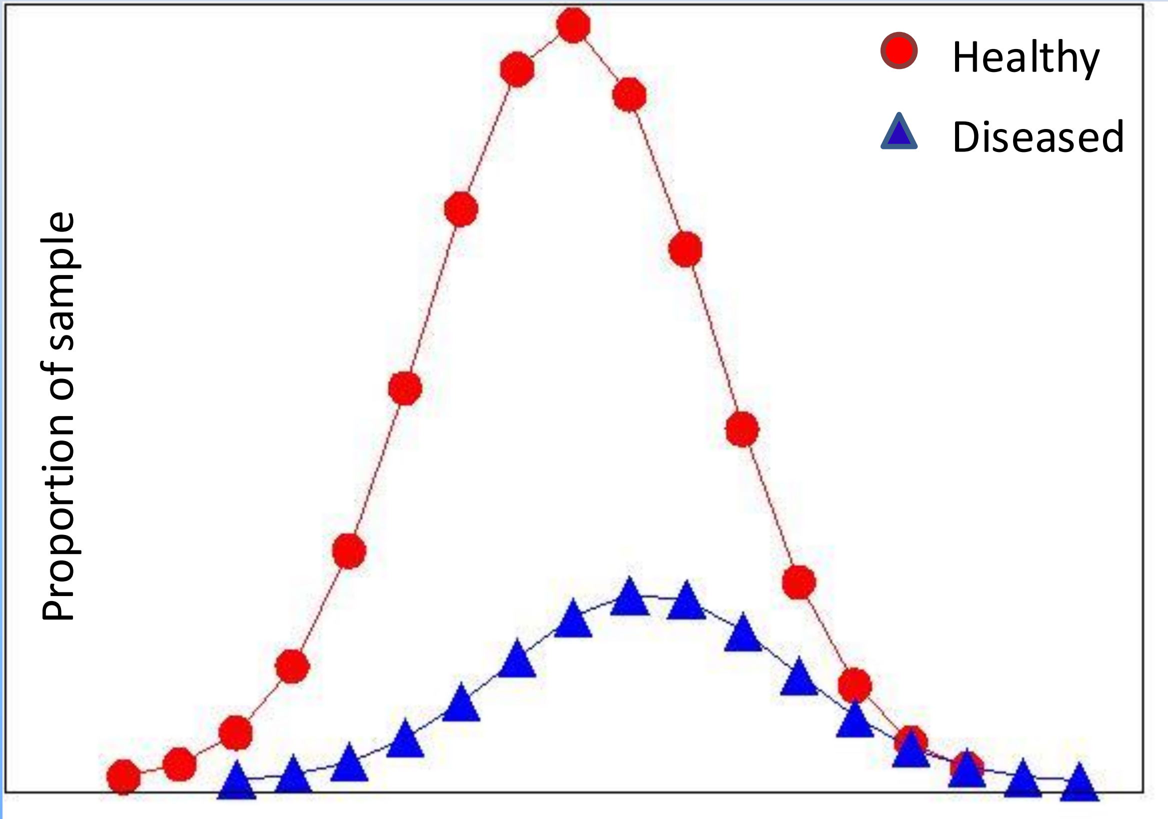
The distribution of the biomarker: which statistical test/method to use for biomarker screening?



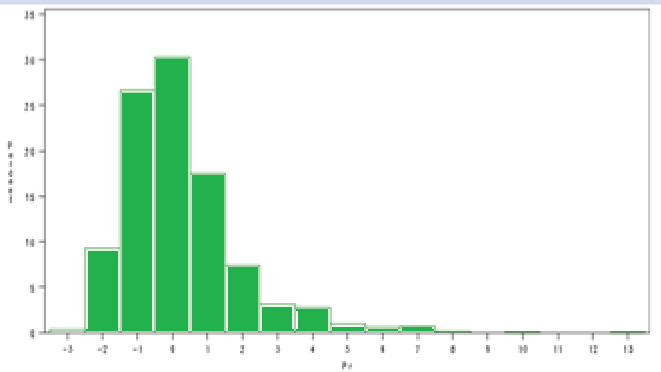
$Mean_{diff}$ = 0.50 SD



\-   Biomarker   +

<u>Common:</u> Evaluate the mean difference
In a biomarker in the "Healthy" versus the "Diseased" using t-test.

The Distribution of the biomarker: Which statistical test/method to use for biomarker screening?



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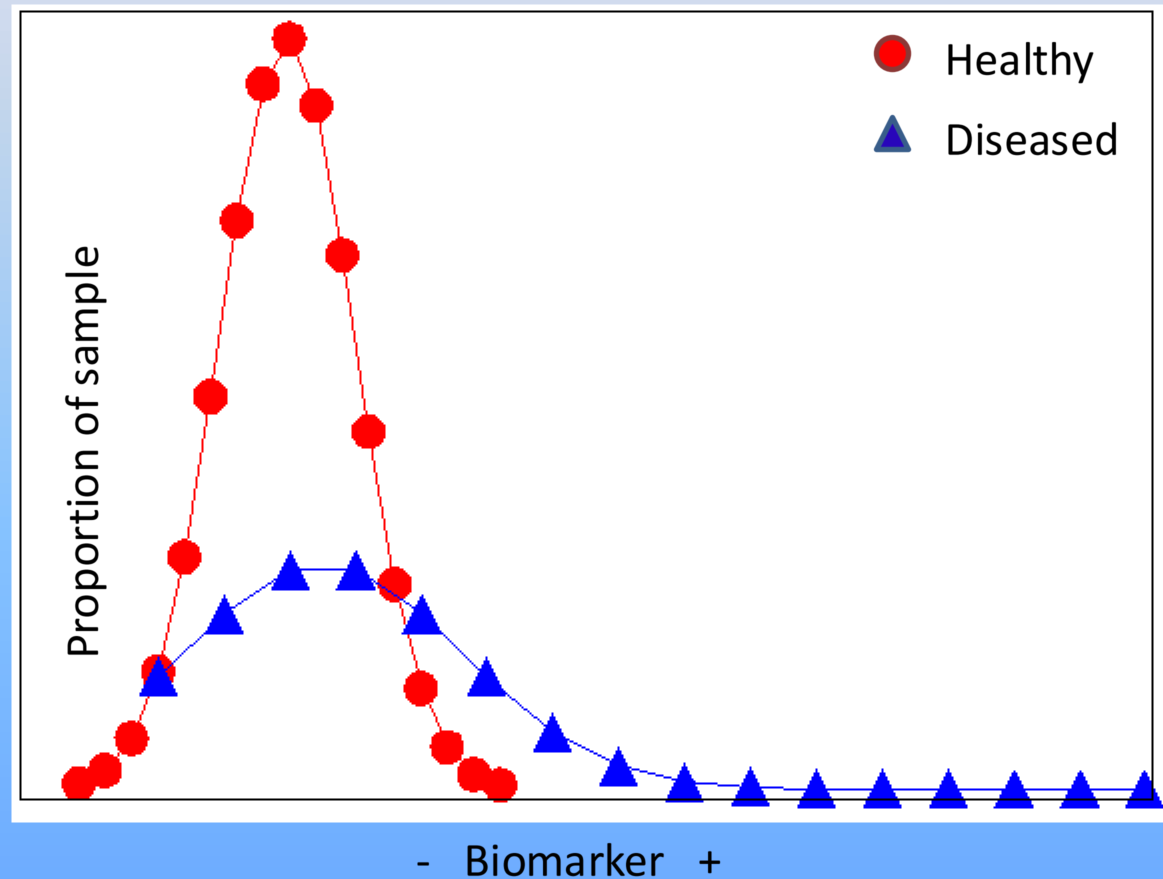
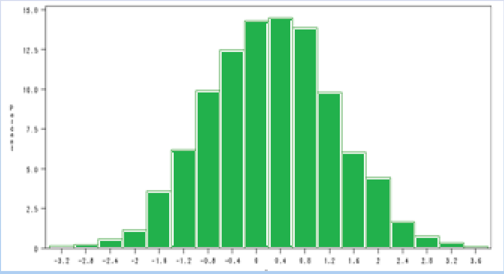
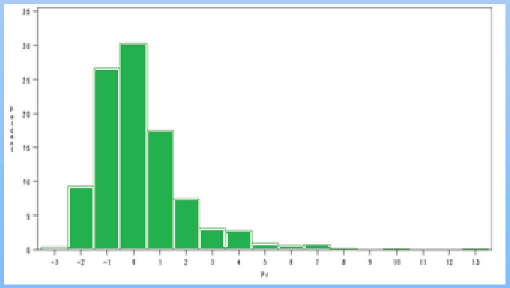


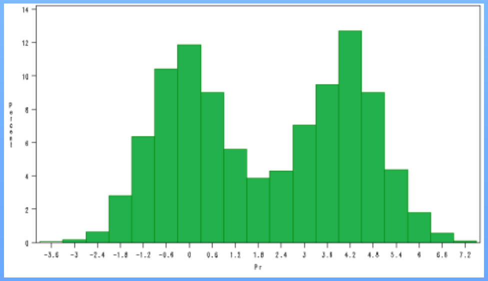
<u>Common:</u> Evaluate the mean difference
In a biomarker in the “Healthy” versus the “Diseased” using Mann-Whitney test.

The Distribution of the biomarker: Which statistical test/method to use for biomarker screening?

- What is the distribution of the biomarker in the population (and sub populations)?
- Often distributions are explored only as a means to the end of applying a significance test
- This is a shame because knowledge of the underlying distributions tells us a lot about the nature of the process of which the biomarker is describing
  - Continua versus categorical







Statistical significance is a necessary but not sufficient descriptor of the utility of a biomarker

Failure to conduct and/or report a statistical power analysis

- Power analyses are very important for interpreting null findings (a positive findings, too).
- Without power, cannot distinguish between the lack of a ‘true’ effect versus the lack of power to detect one
  - Clinically meaningful difference specified
- Post hoc power analyses are misleading

Missing data (or lost data)

- Missing data pose a threat to the interpretation of all studies

- Data not collected for all individuals
    - Only those not in acute distress
    - Only those with a certain hospital length of stay
    - Only those animals that survived experimentation

  – Biomarker assay not available for all levels
    - Minimal detectable threshold

Missing data in submitted manuscripts

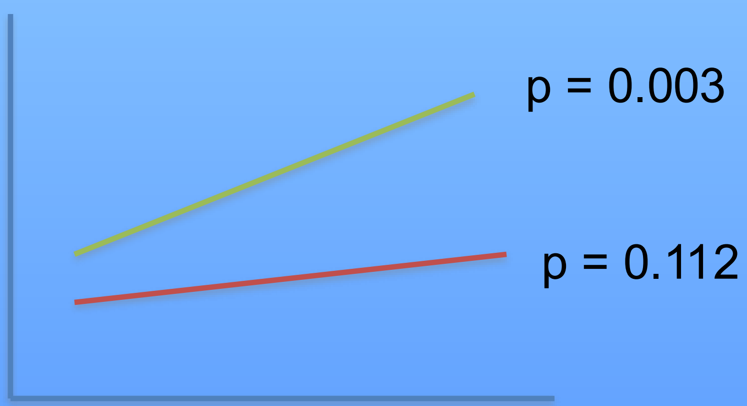
What to do about missing data?

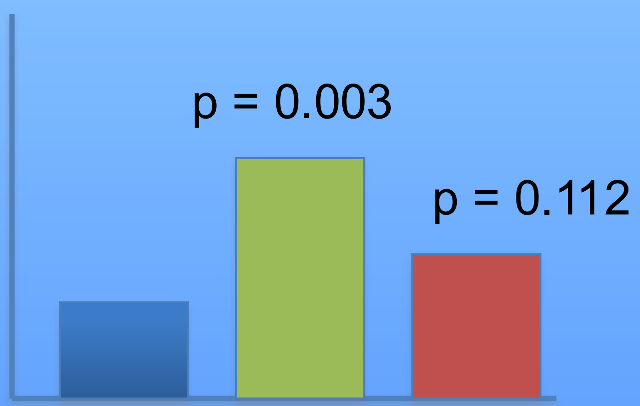
- Must know that data are "missing"
  - Reporting of protocol is crucial (STARD)
    - What patient groups are not being included?
- Characterization of missing data
  - For whom, when, and theoretical effects
  - Standard missing value analysis
- Then, estimation of bias is possible
  - Multiple imputation, clustering methods, etc.
  - Secondary to experimental design (the best way to handle missing data is to not have it)

Comparing p-values as evidence of a differential effect

- Only in rare circumstances can p-values be directly compared to conclude differences in groups

- Differences must be directly tested.





The evaluation of a "straw-man" hypothesis test, statistical power issues

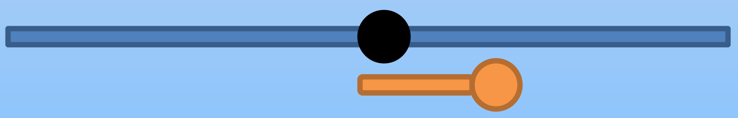
"Straw-man" null hypothesis

0.78 (95% CI 0.74 to 0.82)

Testing sensitivity/specificity
against zero, $p < 0.0001$

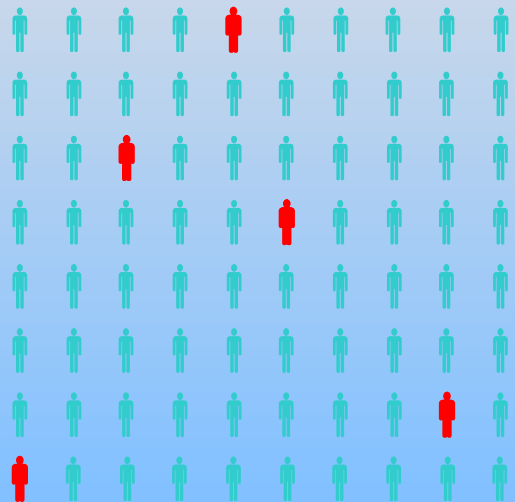
"Interesting" null hypothesis

0.78 (95% LB 0.68)



Testing sensitivity/specificity
against some interesting value (0.70), $p = 0.072$

Problem of “over-fitting”, Type-I error rate



Screening 100 biomarkers
Will likely result in 5 statistically significant associations ($p < 0.05$)

- Most biomarker studies (still) rely on statistical inferences for screening
- If screening is used for a pool of biomarker candidates, control for type-I error should be employed
- Several methods could be used, each with their own strengths and weaknesses
  - Bonferroni adjustments
  - False discovery rates (FDR)
  - More all the time

Statistical significance is a necessary but not sufficient descriptor of the utility of a biomarker

Problem of “over-fitting”, Type-I error rate

- Another problem in assessing biomarkers in the context of a regression model is “over-fitting”
  - Model describes random error instead of underlying relationship
  - The uncovered relationships exploit chance associations in the data

- Occurs when the ratio of predictors (or weights) to outcome is lean, or with multicollinearity (ill-conditioned models)
  - Bias is then expected in the parameter weights
  - The model will not replicate as well as desired

- Can be addressed using a number of simple methods
  - Ample sample size to predictor ratio (e.g., 10:1, 15:1, 30:1)
  - Internal validation techniques (bootstrapping, training/test set)
  - “Phantom” degrees of freedom accounting

Regression to the mean in prediction studies

- A biomarker score can be used to create "extreme" groups from which to evaluate the effectiveness of a treatment using a pre-post test
- Such a design makes regression to the mean a statistical inevitability
  - High scores "get" lower; Low scores "get" higher
  - Occurs whenever there is a non-perfect correlation between the two measurement occasions

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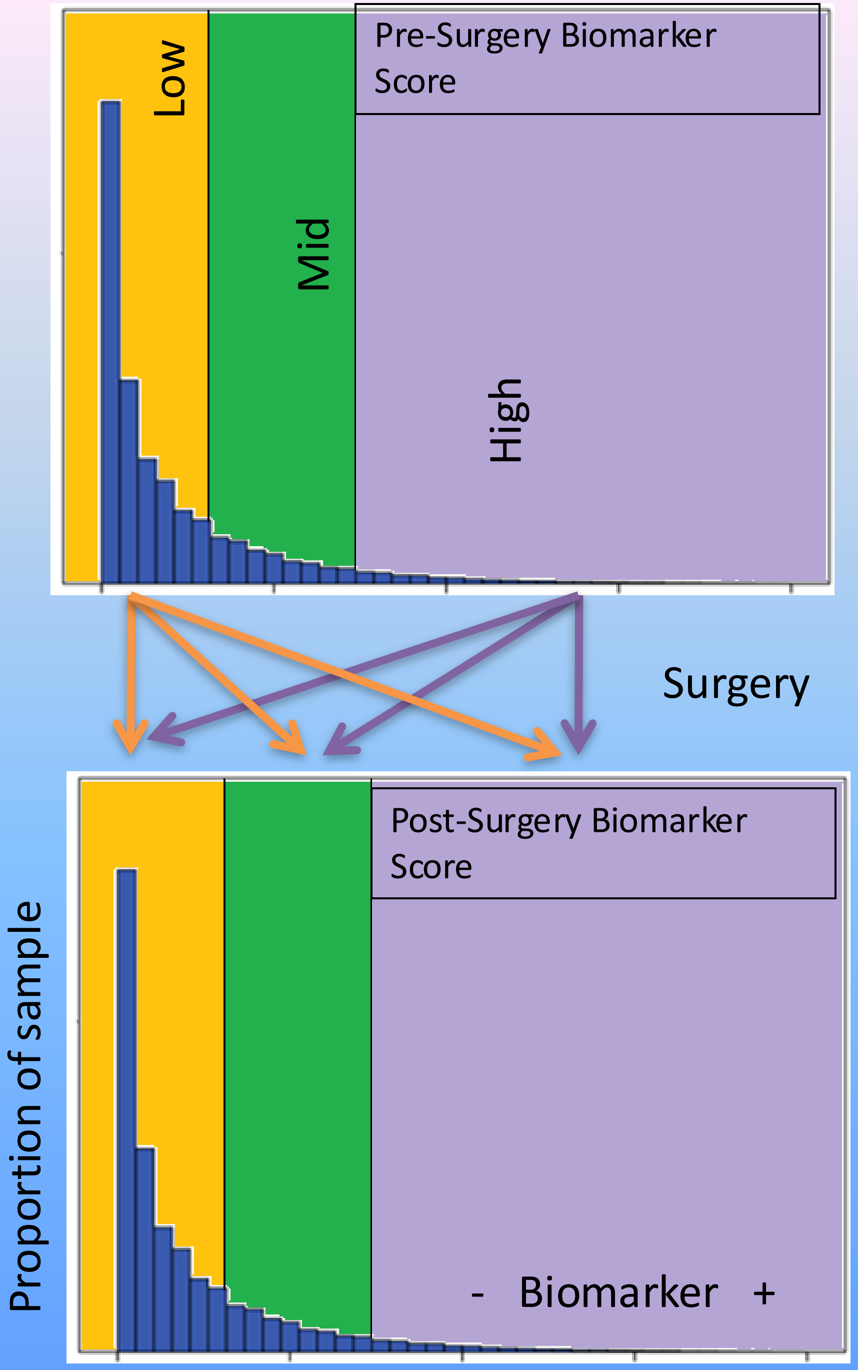
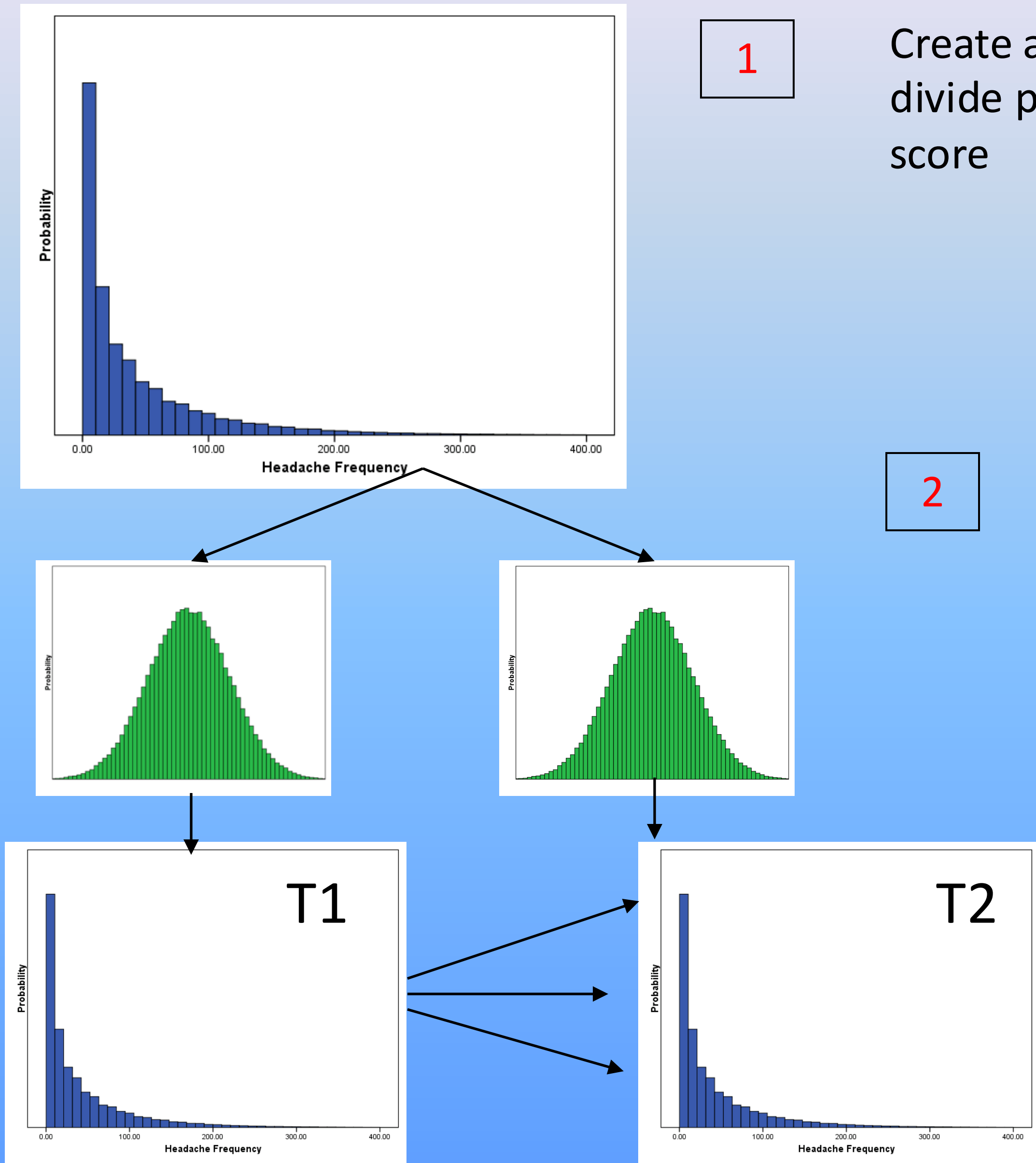


Example

- A team of researchers wishes to examine a group of patients before and after surgery .
- A biomarker score is used to place the patients into one of three groups (low, mid, high)
- The patients are re-assessed after surgery to examine the effect of surgery on the biomarker group
  - How many patients in the high group can be placed in the lower group after surgery?

Simulation Methods



1. Create a "True Score" for a biomarker, divide participants into groups based on score

2. Introduce random variability to create an observed score.

3. Examine changes in biomarker expected through regression to the mean

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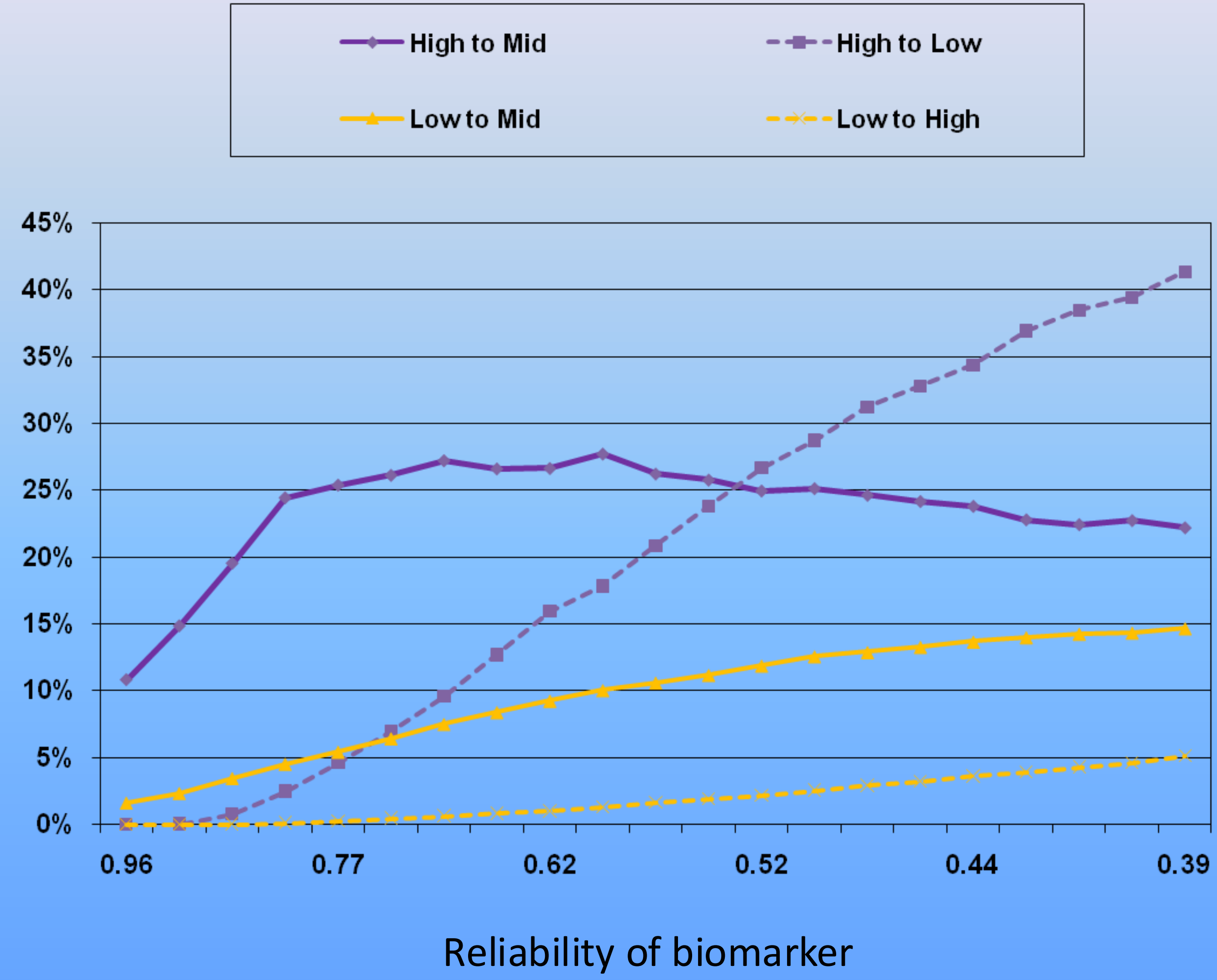
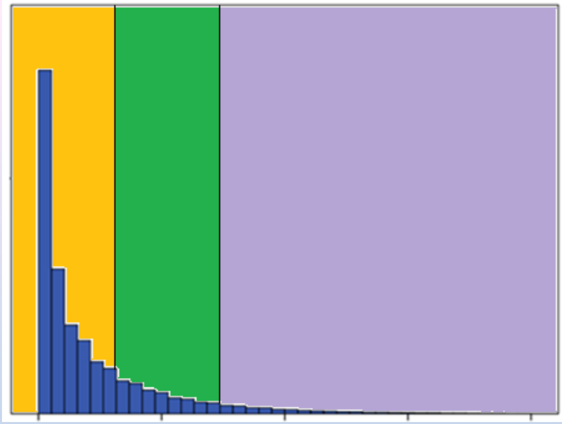
Simulation Results

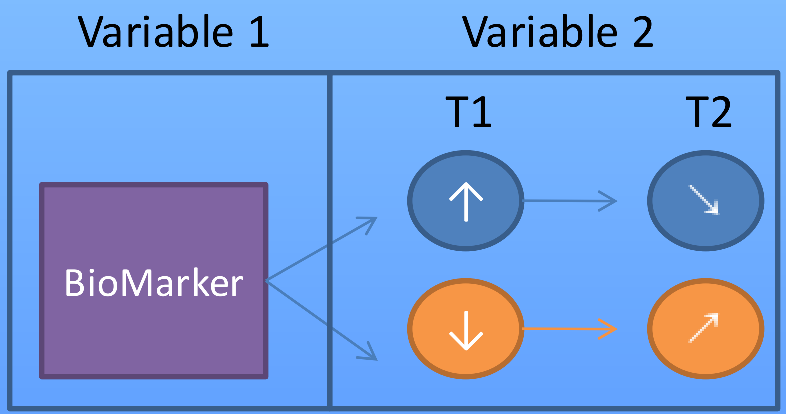




Regression to the mean should be considered in biomarker treatment studies

- The preceding example was perhaps overly-simplistic

- However, regression to the mean should be expected for any study involving change
  - For all non-perfectly reliable biomarkers
  - One variable over time OR one variable used to classify a second variable



Dichotomization of a variable that is naturally continuous

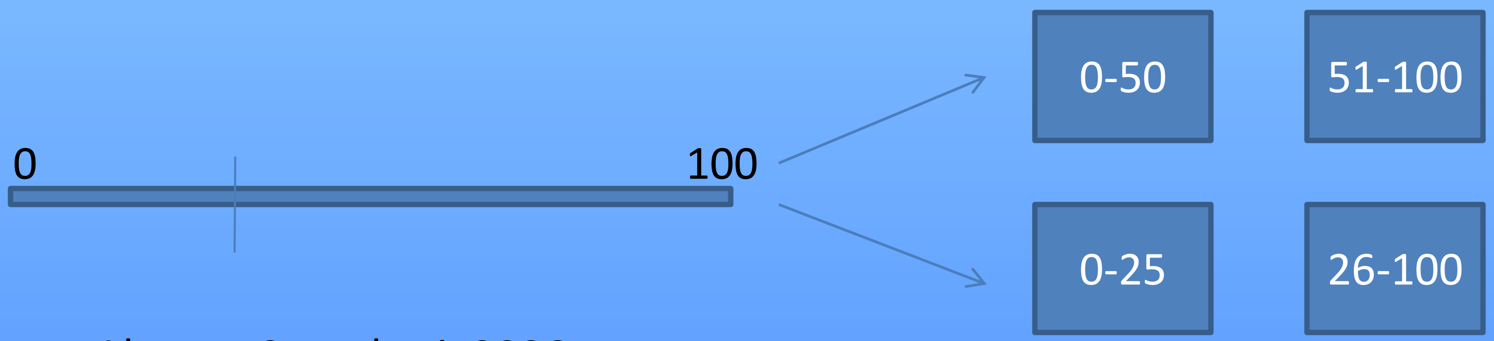
- Although proven to be a poor practice (1993), medical researchers continue to transform continuous variables (e.g., BMI) into discrete categories for analysis (e.g., underweight, normal, obese)

- For clinical purposes this can make sense, as in a decision point

- However for the statistical evaluation of a biomarker, this choice is quite often a bad idea.

Dichotomization of a biomarker that is naturally continuous

- <u>Why do this?</u>
  – Simplify analysis (yes)
  – Simplify interpretation (yes and no)
  – Dichotomizing is more reliable than using imprecise measurements (no)
  – An “optimal” cut-off can be used (yes, at a cost)



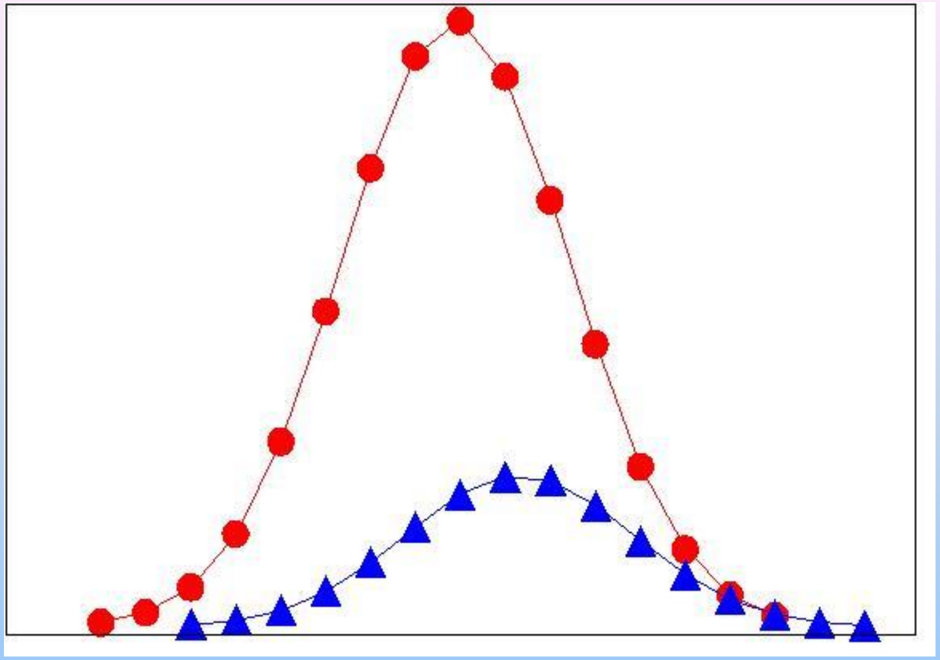
Royston, Altman, Sauerbrei, 2006

Dichotomization of a biomarker that is naturally continuous

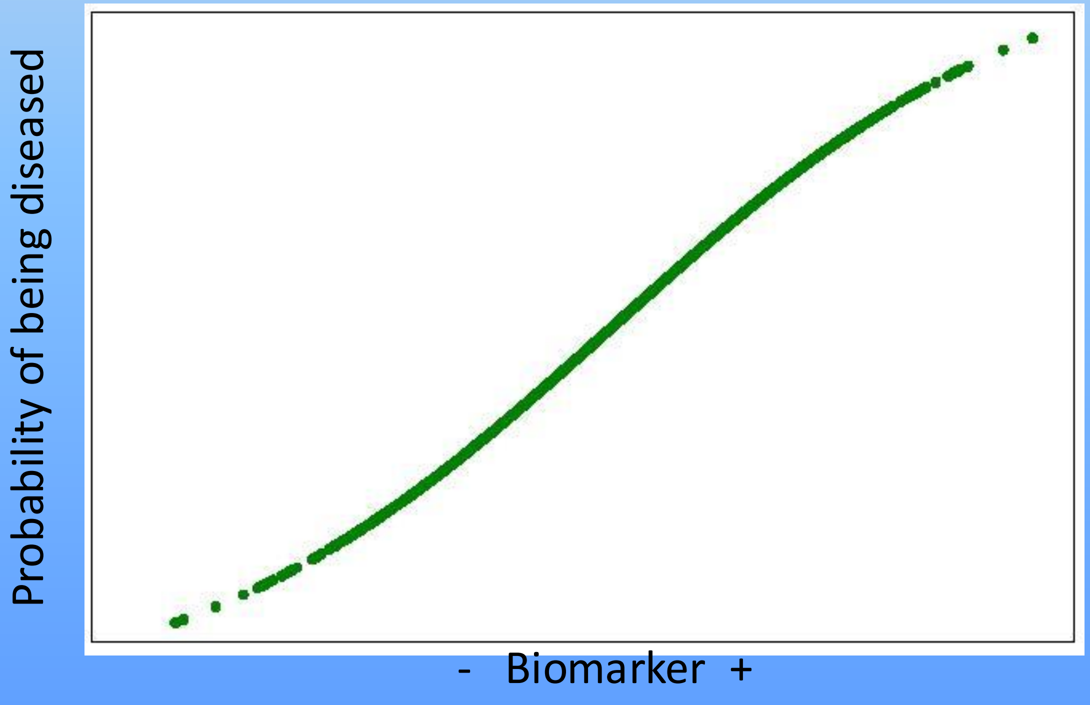
- Why NOT do this?
  - Loss of statistical power
    - When ~N(μ,σ), analyses are only 65% as efficient
      - Might as well throw away 1/3rd of the data!
  - Loss of information
    - Non-linearity cannot be examined
    - Within-group variability absorbed in the entire group
  - Scores close to the cut-point are viewed inappropriately
  - Dichotomized confounders are not well modeled (residual confounding)
  - Ambiguous choice of the "optimal" cut-off

Royston, Altman, Sauerbrei, 2006

2 Dichotomization of a biomarker that is naturally continuous



Difference = 0.50 SD



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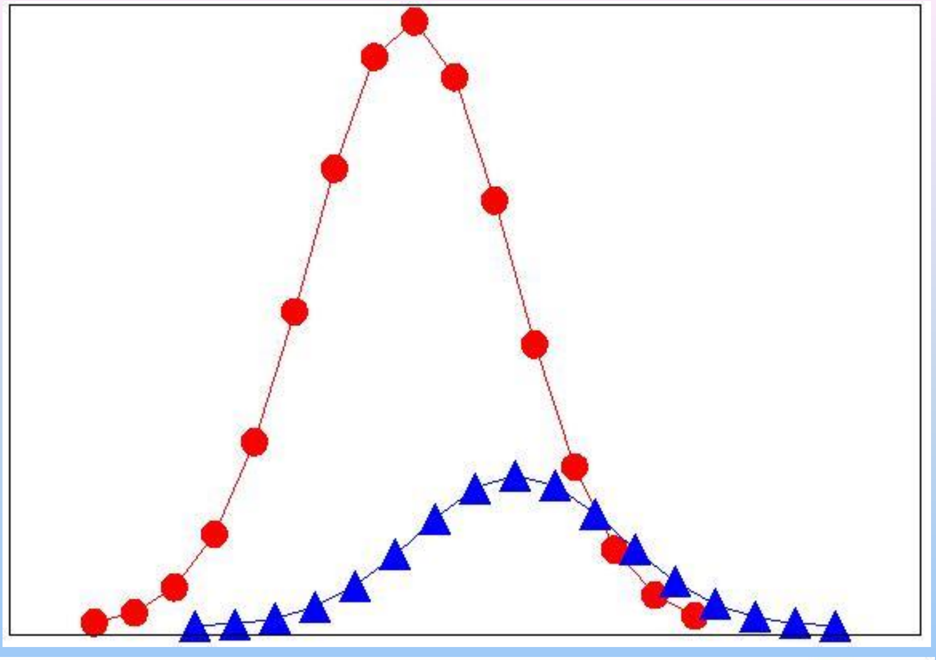
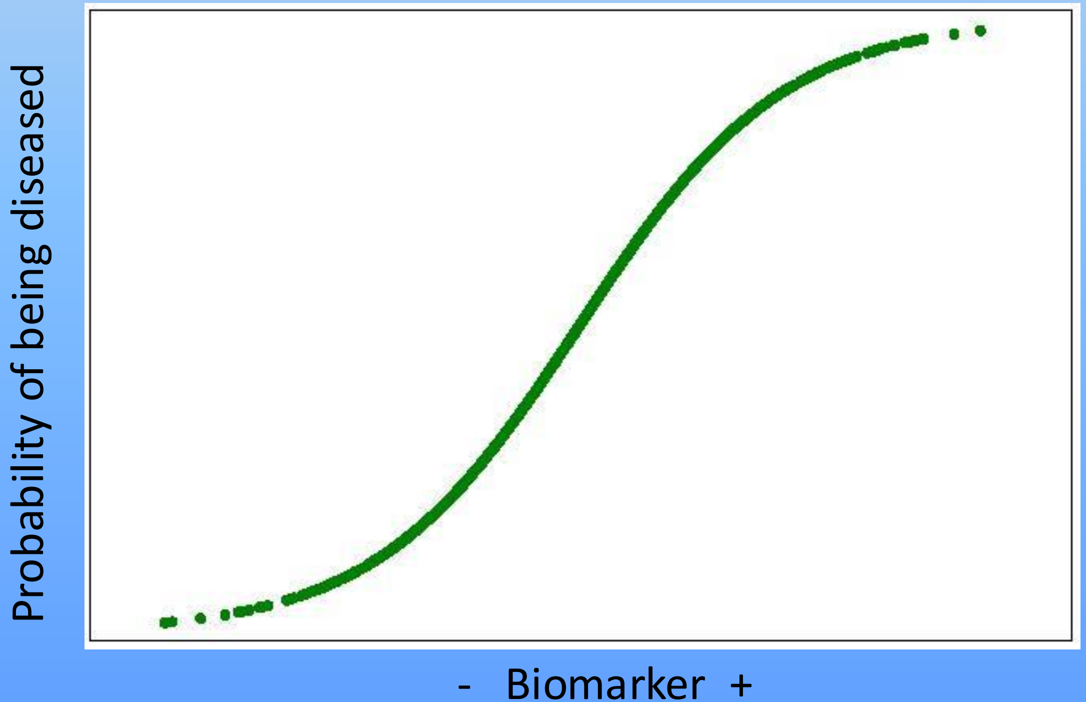
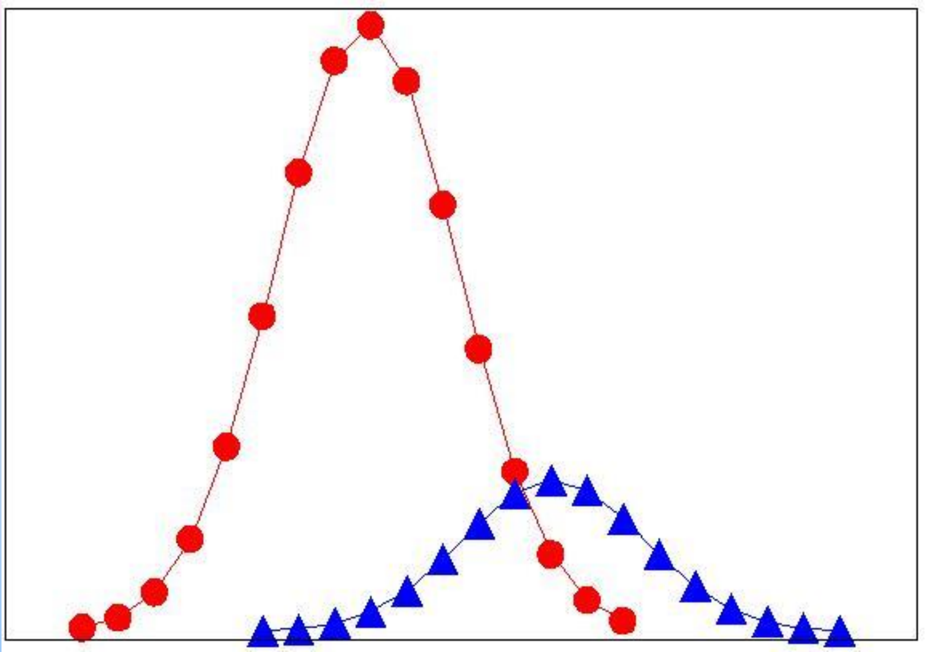


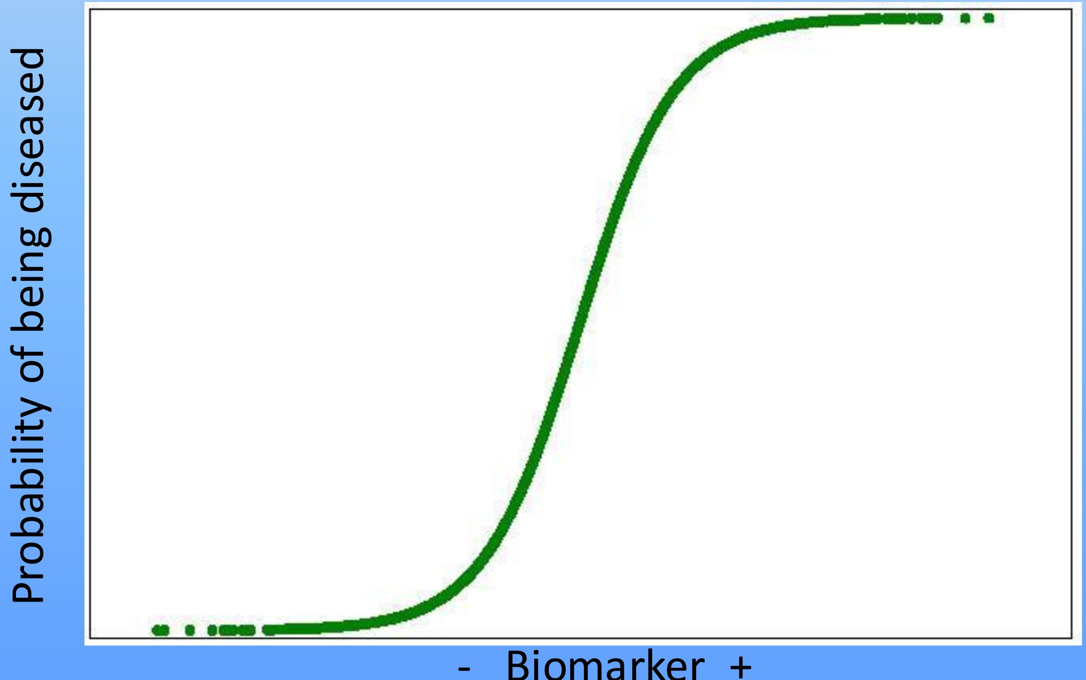
Difference = 1.0 SD



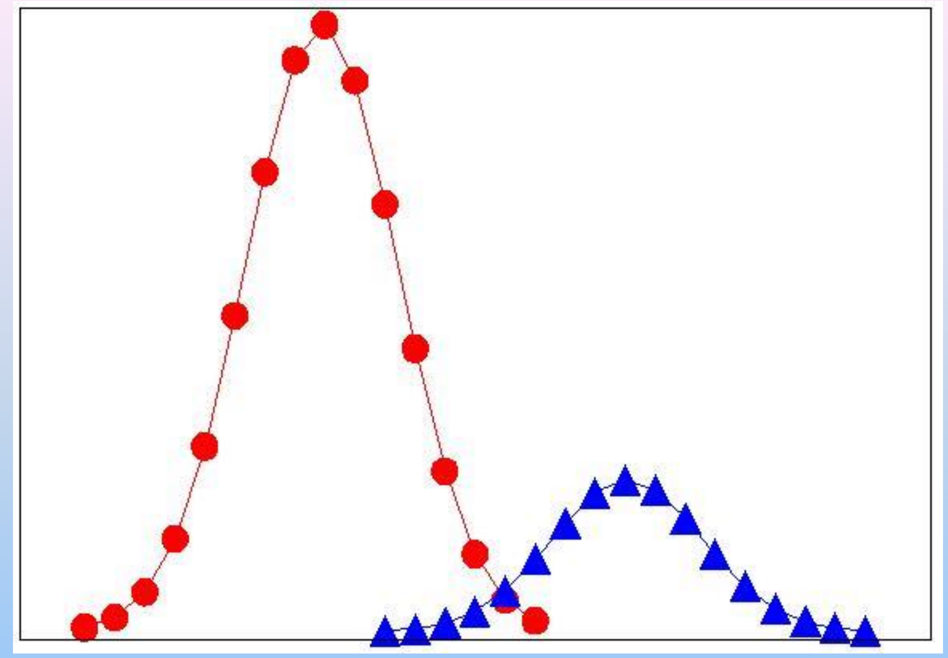
2 Dichotomization of a biomarker that is naturally continuous



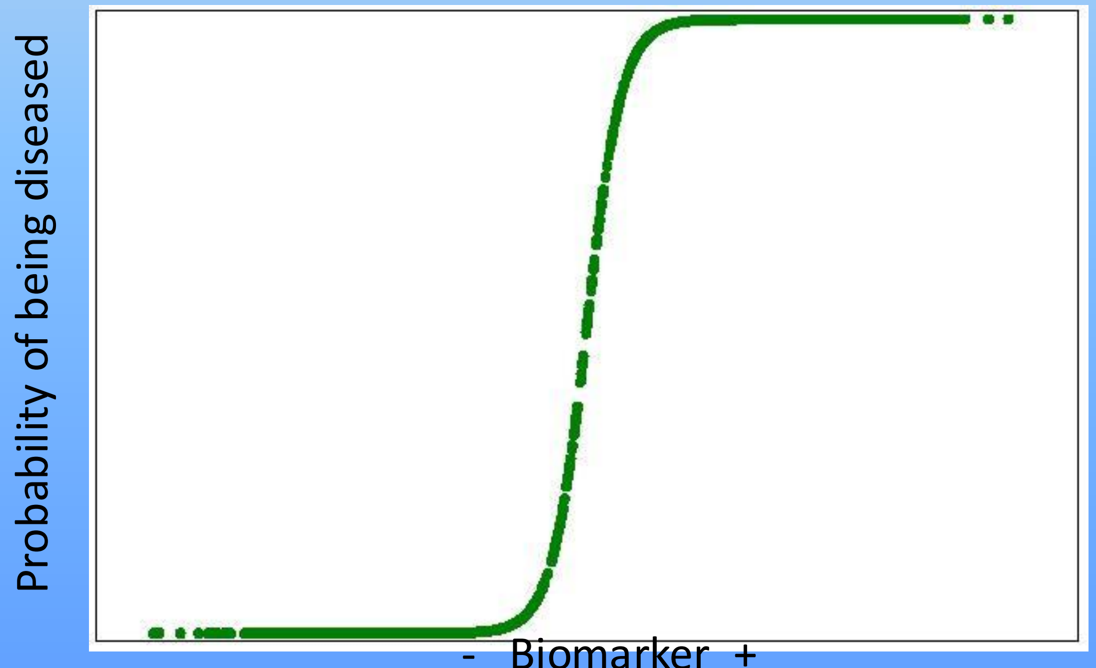
Difference = 2.0 SD



2 Dichotomization of a biomarker that is naturally continuous



Difference = 4.0 SD



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The (un)reliability of measurements

- Studies in medicine seldom explore the statistical reliability of the measurements
  - Perhaps because the variables drawn from a biological sample, it is assumed to have high reliability (i.e., precision).

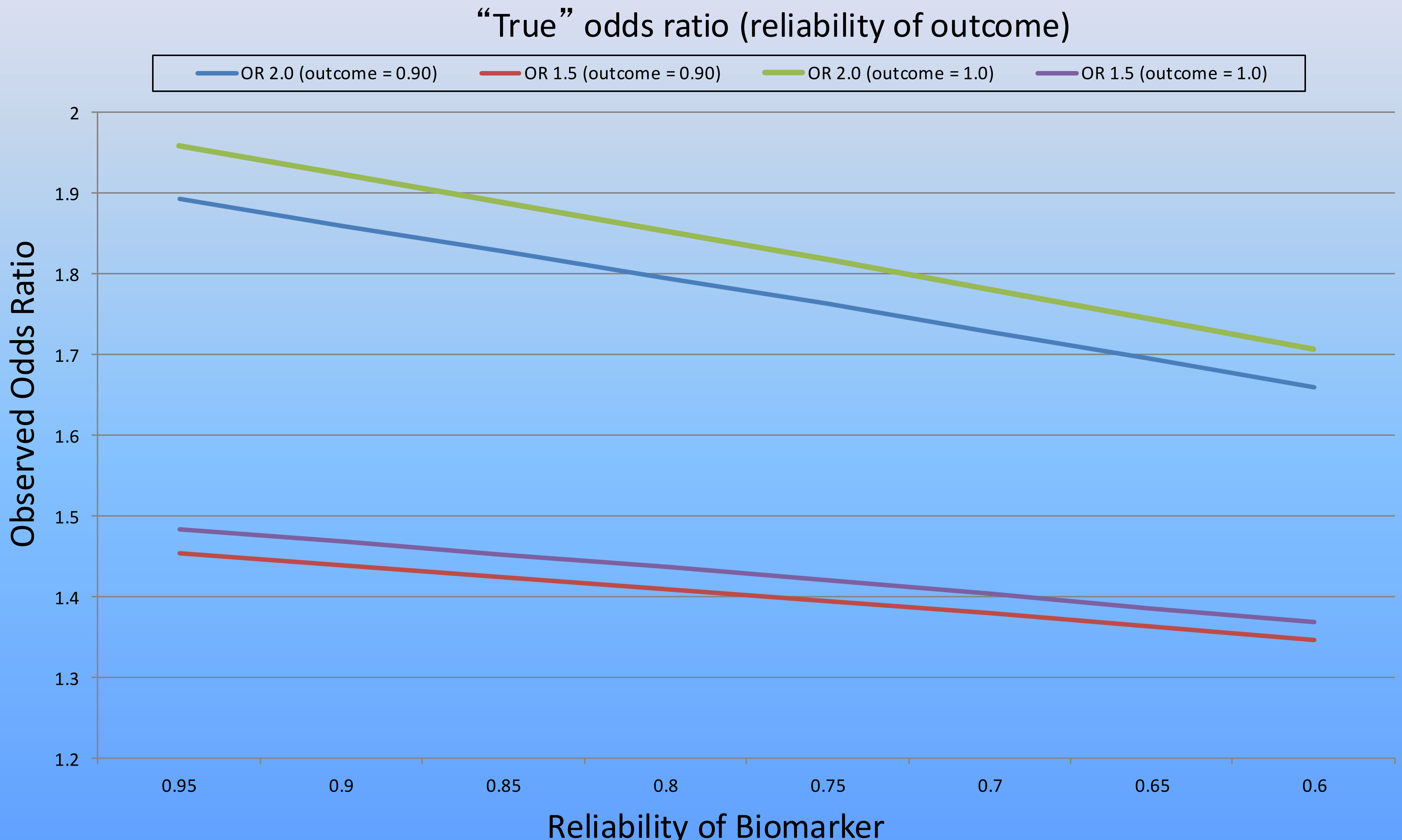
- However, the measurement of these variables can be obscured by error like any other measurement
  - Measurement error (random error)
    - Imprecision of assay,
  - Systematic error (bias)
    - Diurnal variation of a biologic variable

- Reliability and validity are contextual (i.e., for a specific setting)

$$\text{"True" association}_{xy} = \frac{\text{Observed Association}_{xy}}{\text{Reliability}_x * \text{Reliability}_y} \quad = \quad r_{x'y'} = \frac{r_{xy}}{\sqrt{r_{xx} * r_{yy}}}$$

1

The reliability of a biomarker/outcome



Thanks!

1. The reliability of a measurement
2. Artificial dichotomization
3. Regression to the mean
4. “Over-fitting”, Type-I error
5. “Straw-man” hypotheses
6. Comparing p-values
7. Missing data
8. Power analyses
9. Assumptions of tests
10. Lack of a clear analytical plan